School

of

Electronics and Communication Engineering

Minor Project Report

on

Vision-based Techniques to Evaluate the Effectualness of Micro Suturing by Trainee Neurosurgeons

By:

1. **Kiran Siddeshwar** USN: 01FE17BEC080
2. **Varsha Radder** USN: 01FE17BEC213
3. **Pratiksha Danti** USN: 01FE17BCS138
4. **Namrata Nyamagoudar** USN: 01FE17BCS248

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Under the Guidance of

Dr. Uma Mudenagudi

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SCHOOL OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that project entitled **"Vision-based Techniques to Evaluate the Effectualness of Micro Suturing by Trainee Neurosurgeons"** is a bonafide work carried out by the student team consisting of **Kiran K. Siddeshwar (01FE17BEC080), Varsha G. Radder (01FE17BEC213), Pratiksha U. Danti (01FE17BCS138) and Namrata A. Nyamagoudar (01FE17BCS248)**. The project report has been approved as it satisfies the requirements with respect to the mini project work prescribed by the university curriculum for B.E (VI Semester) in the School of Electronics and Communications of KLE Technological University for the academic year 2019-2020.

Dr. Uma Mudenagudi	**Dr. Nalini C. Iyer**	**Prof. N. H. Ayachit**
Guide	**Head of School**	**Registrar**

External Viva:

Name of Examiners **Signature with date**

1.

2.

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-The project team

ABSTRACT

The style of imparting medical training has evolved, over the years. The traditional methods of teaching and practicing basic surgical skills under apprenticeship model, no longer occupy the first place in modern technically demanding advanced surgical disciplines like neurosurgery. Neurosurgery is a challenging surgical specialty that demands many technical and cognitive skills. The traditional surgical training approach of having a trainee coached in the operating room by the faculty is time-consuming, costly, and involves patient risk factors.

Furthermore, the legal and ethical concerns for patient safety as well as cost-effectiveness have forced neurosurgeons to master the necessary micro-surgical techniques to accomplish desired results. This has lead to increased emphasis on assessment of clinical and surgical techniques of the neurosurgeons. A few initiatives using computer-based techniques, have been made to introduce objective evaluation of surgical skills. Due to heavy scarcity of trained neurosurgeons, it is practically impossible to have manual evaluation. This work presents a novel approach involving computerized evaluation of micro-suturing techniques.

In order to train neurosurgeons on Micro Suturing, AIIMS has come up with a training protocol in which they train neurosurgeons to perform suturing on a specially designed sheet/Glove outside the OT. Image processing and computer vision based techniques were then applied to these images to assess the effectualness based on parameters like inter-suture distance, number of sutures per box, slack in the suture, symmetry of the suture and angulation in the suture.

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Chapter 1

Introduction

Neurosurgery is a challenging surgical specialty that demands many technical and cognitive skills. Neurosurgical procedures pose series of challenges for the neurosurgeons, requiring detailed knowledge of micro-neuro-anatomy and well-versed surgical skills. Owing to the complex anatomy, the margin of error in these surgical procedures is very low. Even a minor surgical error made by the surgeon can lead to irreversible catastrophes like paralysis and death. Neurosurgery can be broadly classified into:

- Microscopic surgeries-Surgeries that are performed on very small structures, such as blood vessels and nerves, with specialized instruments under a microscope.
- Endoscopic surgeries-Endoscopic surgeries are performed using a scope, a flexible tube with a camera and light at the tip.
- Fluoroscopic- It is a diagnostic scan used to study the movement of body structures.

The traditional surgical training approach of having a trainee coached in the operating room by the faculty is time-consuming, costly, and involves patient risk factors. Hence, there is a need to have surgical training outside the Operation Theatre (OT). Major skills that a neurosurgeon has to acquire for performing microscopic surgeries are:

1. Micro Suturing
2. Micro Drilling

In order to train neurosurgeons on Micro Suturing, AIIMS has come up with a training protocol in which they train neurosurgeons to perform suturing on a specially designed sheet/Glove outside the OT.

1.1 Motivation

1. Evaluating the performance of the trainee neurosurgeons help them improve their skills.
2. The evaluation can be done on the process (video) or the effectualness (Final image).
3. Evaluating Effectualness is logically more intuitive.
4. Evaluation, if done manually consumes lot of time and resources.

5. Due to heavy scarcity of trained neurosurgeons, it is practically impossible to have manual evaluation.

6. Hence, it is necessary to develop an automatic vision based technique to evaluate the effectualness.

1.2 Objectives

The main objective of the project is to develop a vision based parametric evaluation scheme for micro suturing. The parameters for evaluation include,

- Inter suture distance
- Position of the suture
- Size of the suture
- Angulation in the suture

Other objectives are:

- To keep the system interpretable enough for easy FDA approval
- To keep the system time complexity as low as possible by maintaining conditional independence between the parameters
- To make the system easy to operate by making the task as automatic as possible

1.3 Literature survey

Standard Hough Transforms - The purpose of the technique is to find imperfect instances of objects within a certain class of shapes by a voting procedure. The classical Hough transform was concerned with the identification of lines in the image. This could be used to detect sutures in the image.

Probabilistic Hough Transforms - This is an efficient implementation of hough transform which could also be used to detect the sutures. It takes a random subset of points that are sufficient for line detection.

Contour Hunting - Contour hunting is one of many preprocessing techniques performed on digital images in order to extract information about their general shape. Once the contour of a given pattern is extracted, it's different characteristics will be examined and used as features which will later on be used in pattern classification. Therefore, correct extraction of the contour will produce more accurate features which will increase the chances of correctly segmenting a given pattern.

Various thresholding techniques - Thresholding is used to segment an image by setting all pixels whose intensity values are above a threshold to a foreground value and all the remaining pixels to a background value.

Adaptive Thresholding - Adaptive thresholding typically takes a grayscale or color image as input and, in the simplest implementation, outputs a binary image representing the segmentation. For each pixel in the image, a threshold is calculated. If the pixel value is below the threshold it is set to the background value, otherwise it assumes the foreground value. By calculating the threshold within the neighborhood area of the image, we can achieve a better result from images with varying illumination.

Viola Jones-Viola-Jones algorithm, [3] is an object-recognition framework that allows the detection of image features in real-time. It focuses on Haar basis feature filters for face detection. The efficiency of the Viola-Jones algorithm can be significantly increased by first generating the integral image.

Haar Cascade Classifier-Haar Cascade is a machine learning object detection algorithm used to identify objects in an image or video and based on the concept of features proposed by [3]. LBP, Local Binary Pattern operator [1], the idea behind this operator is that common features, such as edges, lines, point, can be represented by a value in a particular numerical scale. Therefore, it is possible to recognize objects in an image using a set of values extracted a priory.

HOG-SVM Classifier-Support vector machines (SVMs) [2] are supervised machine learning models that divide and classify data. An SVM model can be considered as a point space wherein multiple classes are isolated using hyperplanes. A HOG is a feature descriptor generally used for object detection. A HOG relies on the property of objects within an image to possess the distribution of intensity gradients or edge directions. Given the HOG features x of a window in an image, the SVM assigns a score, which determines how certain the algorithm is that this object is a suture or not (binary classification).

1.4 Problem statement

To develop vision-based techniques to evaluate the effectualness of micro-suturing by trainee neurosurgeons.

1.5 Organization of the report

In chapter 2 System design, we discuss the various method to design the system. In chapter 3 Implementation details, we discuss the detailed implementation of all the experiments. In chapter 4 Results and discussions, we discuss the results of all the experiments. In chapter 5 Conclusions and future scope, we discuss the improvement that can be brought to the final design and the future scope of the project.

Chapter 2

System design

2.1 Functional block diagram

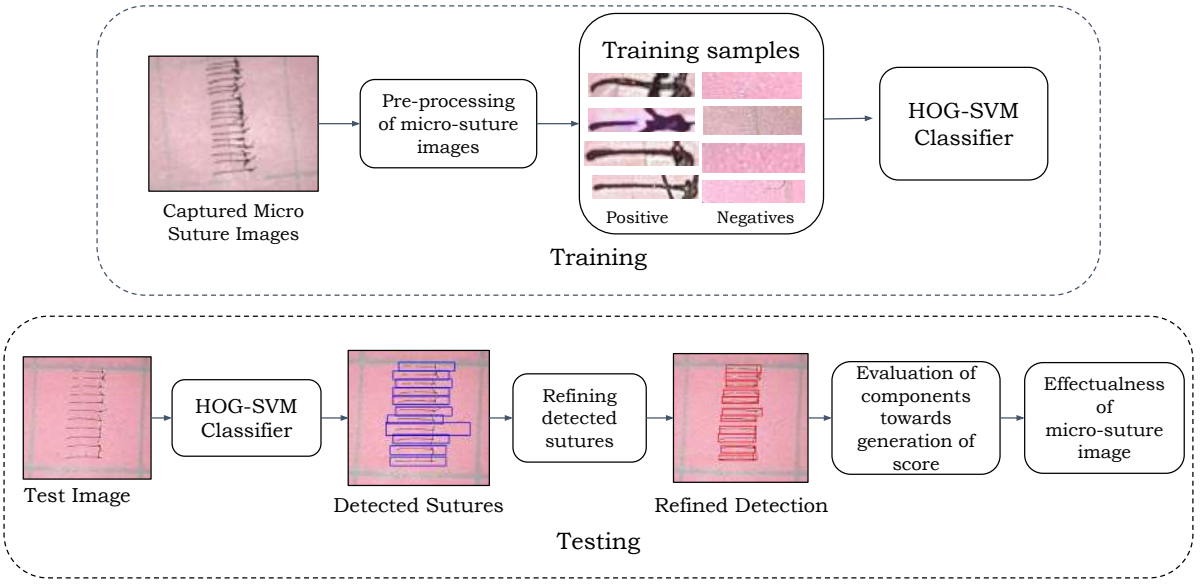


Figure 2.1: Functional Block Diagram

Figure 2.1 shows the pipeline of the proposed system. Initially, the dataset is split into train and test sets. In the training phase, the captured micro-suture image is pre-processed and training samples are generated. A HOG-SVM classifier is trained using these samples. In the testing phase, when an image is fed to the HOG-SVM classifier, sutures are detected. The bounding boxes are further refined using thresholding. This is then used for evaluation of components towards generation of score. A combined score is generated towards demonstration of effectualness of suturing. The final output is the effectualness of micro-suture.

Chapter 3

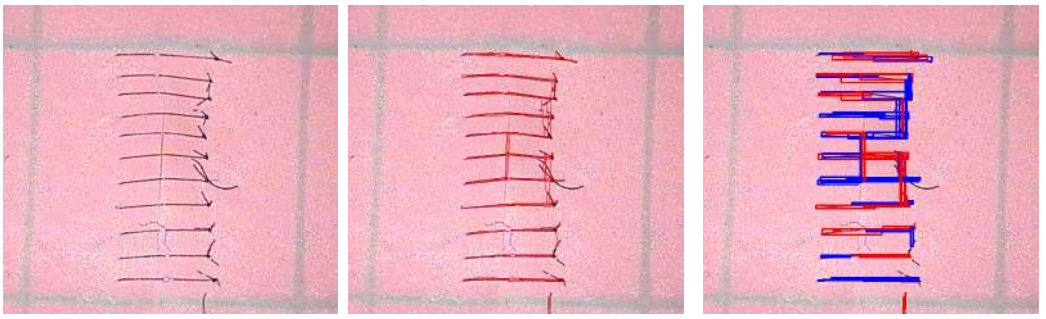
Implementation details

As accurate suture detection and localization is an important step for parametric evaluation, different experimentation have been carried out in-order to perform suture detection.

3.1 Experiments

3.1.1 Probabilistic Hough Transform

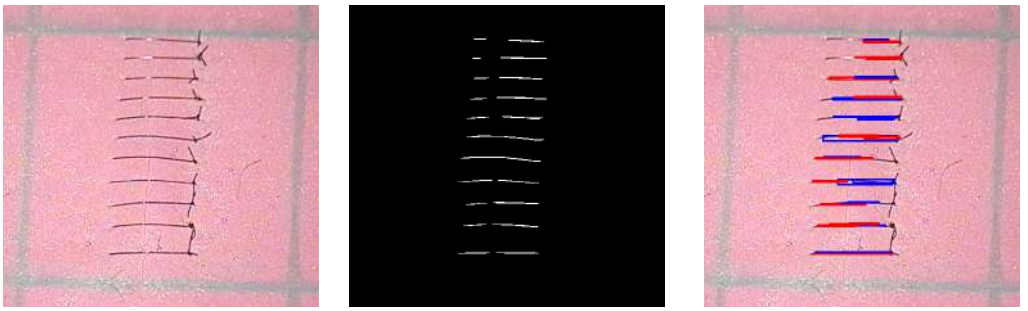
- Probabilistic Hough Transform is an optimization of Hough Transform. Hough Transform is a popular technique to detect any shape, if you can represent that shape in mathematical form.
- The purpose of the technique is to find imperfect instances of objects within a certain class of shapes by a voting procedure.
- The classical Hough transform was concerned with the identification of lines in the image. This could be used to detect sutures in the image. It takes a random subset of points that are sufficient for line detection.



(a) Input (b) After applying PHough Transform (c) Output

3.1.2 Adaptive thresholding and Probabilistic Hough Transform

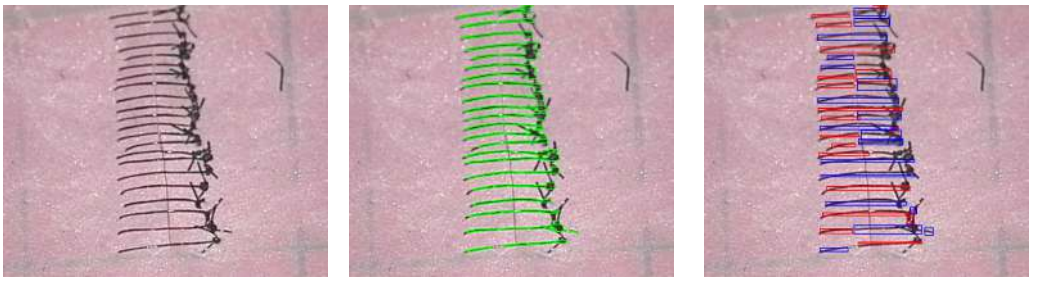
- Adaptive thresholding is the method where the threshold value is calculated for smaller regions and therefore, there will be different threshold values for different regions.
- By using vertical filter on 3.4a, we can get rid of the knots as shown in 3.4b, so that we get only the suture as required.
- Fig. 3.4c is achieved by subtracting the original image from the vertical filtered image, and by applying Hough Transform.



(a) Input (b) After applying vertical filter (c) Output

3.1.3 Contours

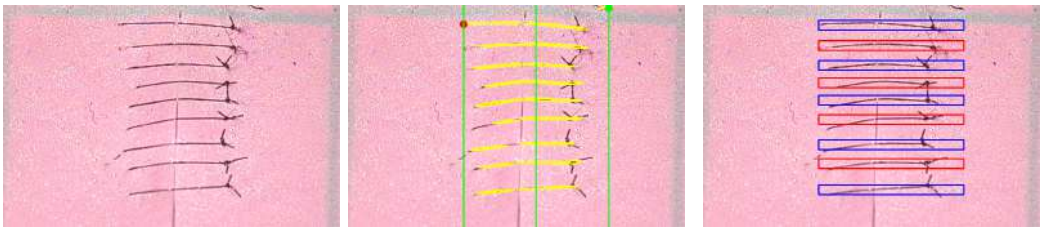
- Contours are curves joining all the continuous points along the boundary having same color or intensity.This helps us to have a way to segment out an object and find its boundary.
- Fig. 3.5b is achieved by applying contours on Fig. 3.8a i.e, the input image.
- Fig. 3.8c is obtained by drawing bounding boxes by taking minimum and maximum coordinates of the contours.



(a) Input (b) After applying Contours (c) Output

3.1.4 Half Contours

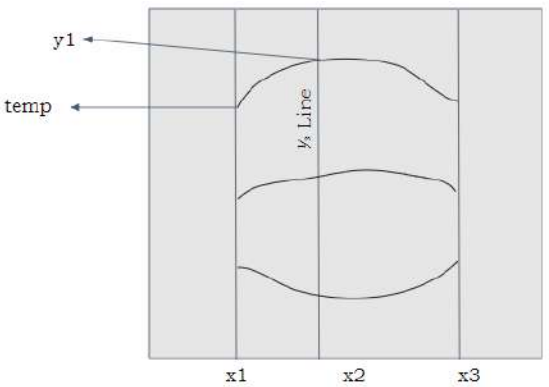
- Initially, the extreme coordinates of the contours are detected and two vertical lines are drawn at these coordinates.
- We draw another vertical line at the point by taking average of the extreme coordinates of the contours as shown in Fig. 3.5b.
- Later we find the intersection between the contours and the one half line.
- Later the bounding boxes are drawn and the height is determined by by taking 5 pixels above the intersection point and 15 pixel below the intersection point as shown in as shown in Fig. 3.5b.



(a) Input (b) (c) Output

3.1.5 One - third Contours

- Initially, the extreme points are detected and two vertical lines are draw at these points.
- We draw another vertical line at one third of the two lines.
- Considering only the left part of the one third vertical line, contours are drawn.
- Using the extreme points of each suture and the below formula we draw the bounding box.



Listing 3.1: Algorithm

```
if (temp>y1):
    y3 = temp
    y2 = y1 - 1/3(y3-y1)
if(temp<y1):
    y2 = temp
    y3 = y1 + 1/3(y2-y1)
temp:extreme left y coordinate
y1:extreme right y coordinate
y2 and y3 are calculated to draw BB
```

(a) Input

(b) Contours for one third part

(c) Output

3.1.6 Haar Cascade

- Haar Cascade is a machine learning object detection algorithm used to identify objects in an image or video and based on the concept of features proposed by Paul Viola and Michael Jones in their paper "Rapid Object Detection using a Boosted Cascade of Simple Features".

- The cascade classifier consists of a collection of stages, where each stage is an ensemble of weak learners. The weak learners are simple classifiers called decision stumps. Each stage is trained using a technique called boosting. Boosting provides the ability to train a highly accurate classifier by taking a weighted average of the decisions made by the weak learners

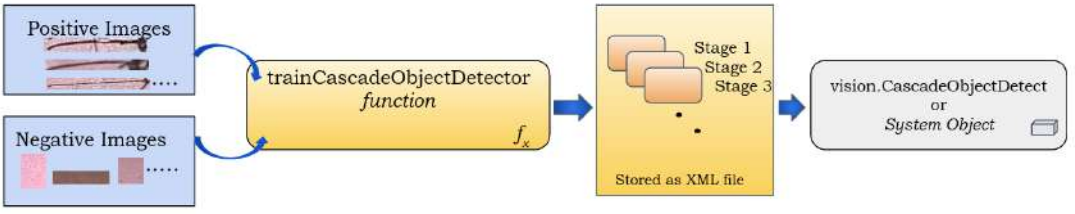


Figure 3.6: Haar Cascade

- We used LBP feature type [1] for training the cascade object detector in order to obtain better results

- LBP, Local Binary Pattern operator,the idea behind this operator is that common features, such as edges, lines, point, can be represented by a value in a particular numerical scale. Therefore, it is possible to recognize objects in an image using a set of values extracted a priori.

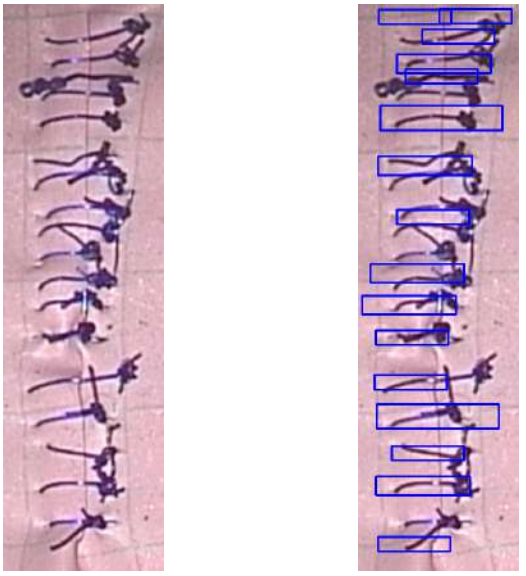
- The original LBP labels the pixels by thresholding the 3x3 neighborhood in relation to the central pixel value, as shown in the equation

$$LBP(x_p, y_p) = \sum_{n=0}^{7} s(i_n - i_p)2^n,$$

  where (x_p, y_p) is the pixel of an image,n represents the neighboring pixel, i_n and i_p the respective gray level of neighboring and central pixel, and $s(x)$ can be described by:

$$s(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- The bit obtained for each neighboring pixel is run through a pre-defined order to obtain a final array of bit values. Using eight neighboring pixels, the final value will be between 0 and 255 for the above 3x3 kernel.



(a) Input (b) Output

3.1.7 HOG-SVM

- To classify a window with a feature vector x, an SVM computes the following function

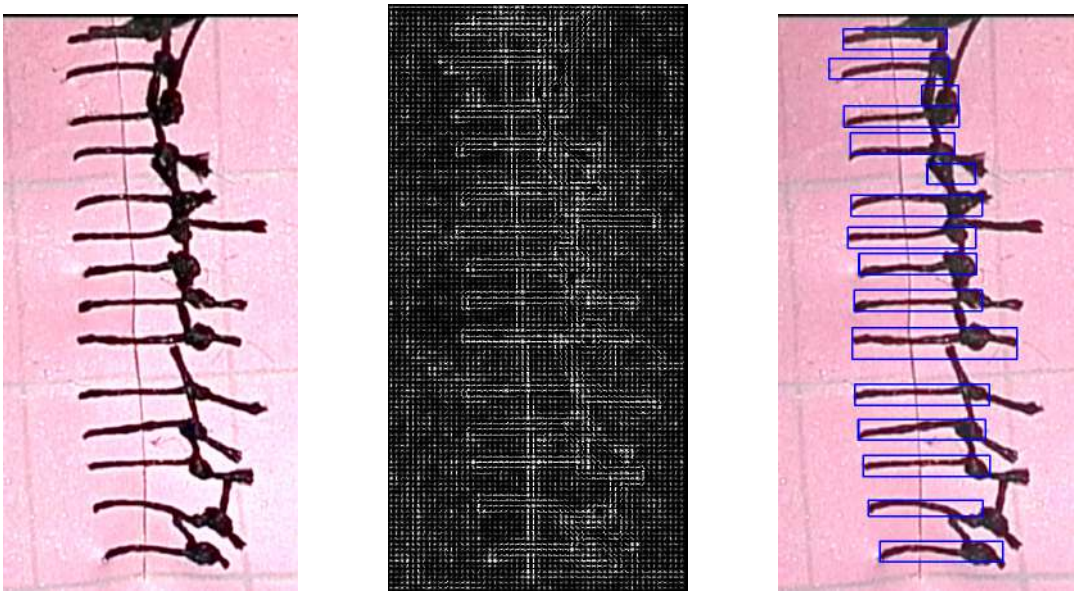
$$h(x) = w^T x + b$$

where w is called the weight vector and b is the bias.

- These two parameters give rise to a hyperplane in feature space, which separates windows containing sutures from background windows in a sliding-window-fashion. A binary decision is commonly achieved by using the sign function on the output of the SVM

$$g(x) = sign(h(x)) = sign(w^T x + b)$$

with $g(x) \in$\{-1,1\}.



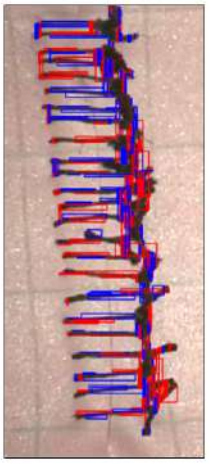
(a) Input (b) HOG Visualization (c) Output

Chapter 4

Results and discussions

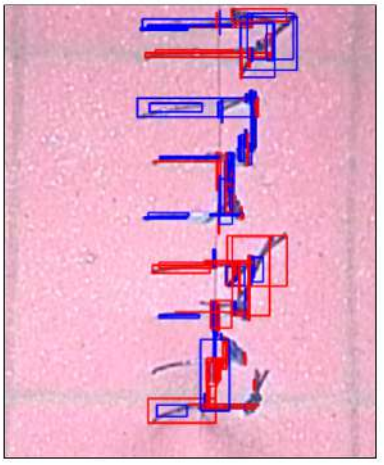
4.1 Result Analysis

4.1.1 Probabilistic Hough Transform

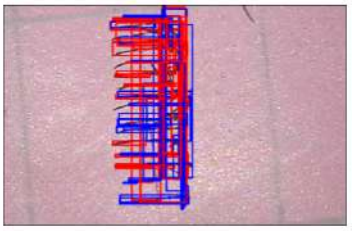


Magnification: 4

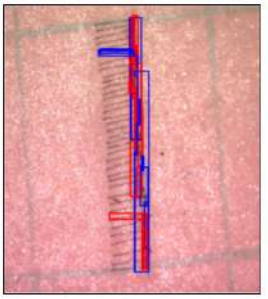
Magnification: 5



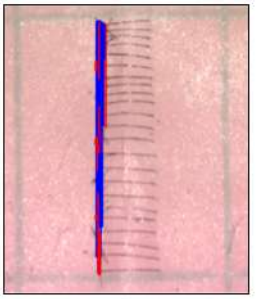
Magnification: 7



Magnification: 8



Magnification: 10



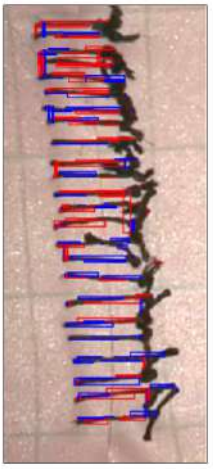
Magnification: 10

mAP for IoU of 0.3 for PHough Transform					
Full Data-set	0.07%	For Mag of 4	0.00%	Bad Good	0.00% 0.00%
		For Mag of 5	0.03%	Bad Good	0.06% 0.00%
		For Mag of 7	0.02%	Bad Good	0.04% 0.01%
		For Mag of 8	0.00%	Bad Good	0.0% 0.00%
		For Mag of 10	0.37%	Bad Good	0.64% 0.11%

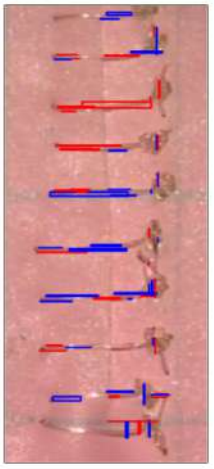
Table 4.1: Results of Probabilistic Hough Transform

4.1.2 Adaptive Thresholding and Hough Transform

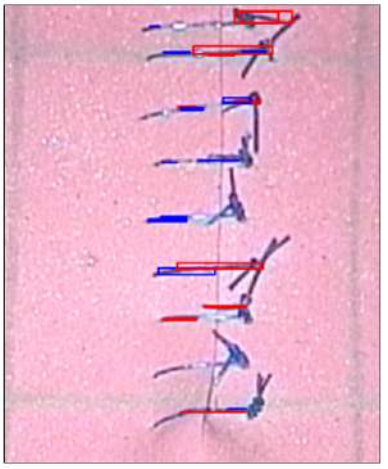
The results of Adaptive Thresholding and Hough Transform with respective magnification level and mAP are as follows:



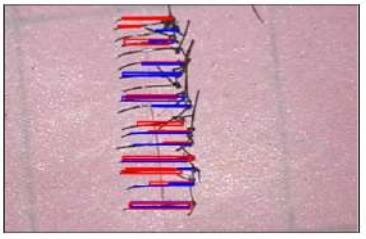
Magnification: 4



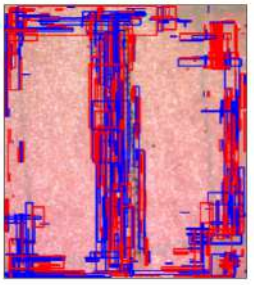
Magnification: 5



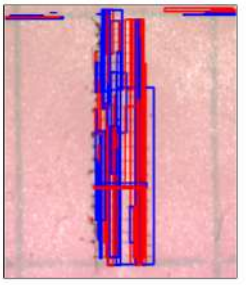
Magnification: 7



Magnification: 8



Magnification: 10



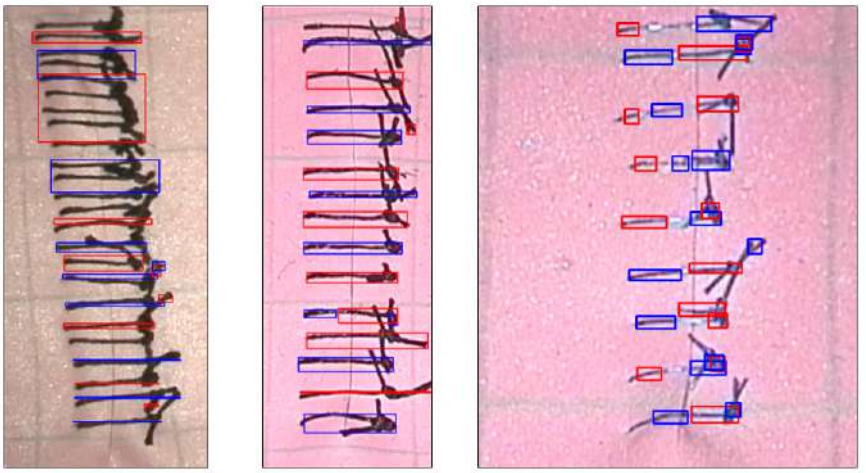
Magnification: 10

mAP for IoU of 0.3 for AT and Hough Transform					
Full Data-set	0.05%	For Mag of 4	0.01%	Bad Good	0.01% 0.13%
		For Mag of 5	0.04%	Bad Good	0.09% 0.00%
		For Mag of 7	0.00%	Bad Good	0.00% 0.00%
		For Mag of 8	0.05%	Bad Good	0.10% 0.04%
		For Mag of 10	0.19%	Bad Good	0.06% 0.83%

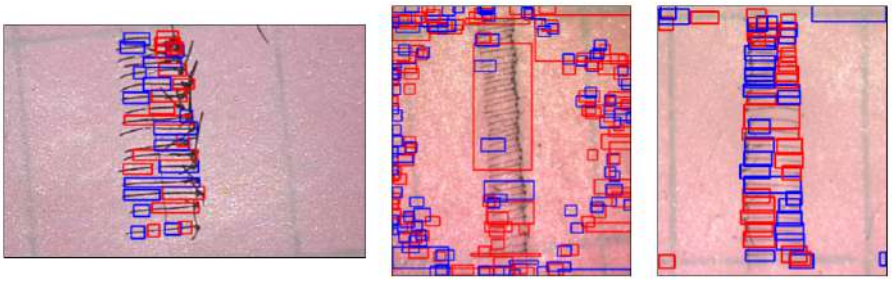
Table 4.2: Results of Adaptive Thresholding and Hough Transform

4.1.3 Contours

The results of Contours with respective magnification level and mAP are as follows:



Magnification: 4 Magnification: 5 Magnification: 7



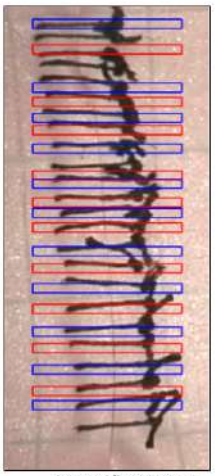
Magnification: 8 Magnification: 10 Magnification: 10

mAP for IoU of 0.3 for Contours					
Full Data-set	4.58%	For Mag of 4	2.26%	Bad Good	2.59% 1.96%
		For Mag of 5	1.68%	Bad Good	2.89% 0.75%
		For Mag of 7	4.22%	Bad Good	6.90% 4.31%
		For Mag of 8	8.95%	Bad Good	17.92% 1.31%
		For Mag of 10	10.48%	Bad Good	2.60% 34.82%

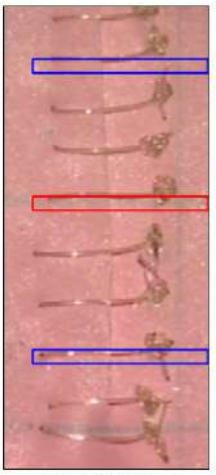
Table 4.3: Results of Contours

4.1.4 Half-Contours

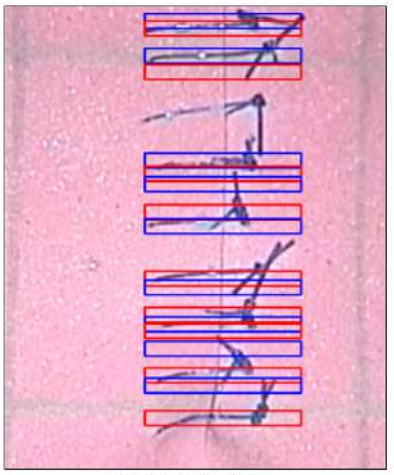
The results of Half-Contours with respective magnification level and mAP are as follows:



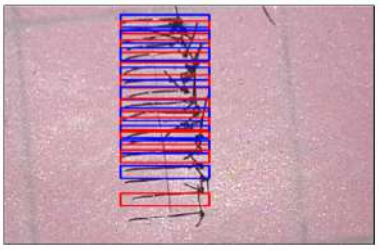
Magnification: 4



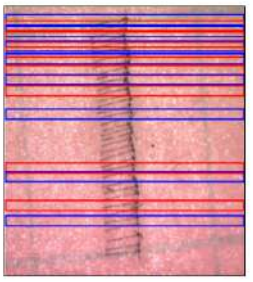
Magnification: 5



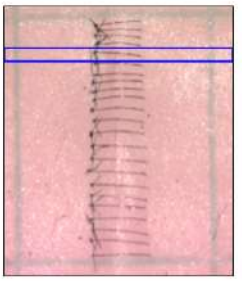
Magnification: 7



Magnification: 8



Magnification: 10



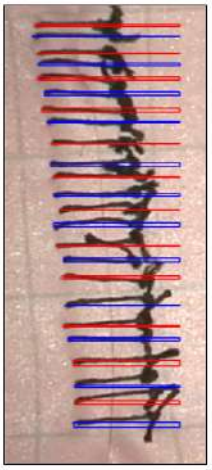
Magnification: 10

mAP for IoU of 0.3 for Half-Contours					
Full Data-set	12.33%	For Mag of 4	13.96%	Bad Good	15.16% 18.05%
		For Mag of 5	16.28%	Bad Good	8.77% 36.70%
		For Mag of 7	34.31%	Bad Good	44.31% 33.24%
		For Mag of 8	16.11%	Bad Good	35.51% 0.00%
		For Mag of 10	1.60%	Bad Good	0.00% 11.18%

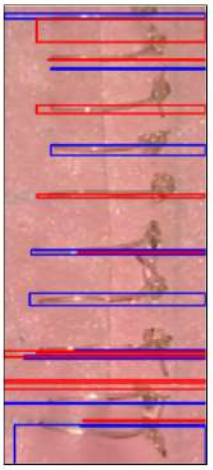
Table 4.4: Results of Half-Contours

4.1.5 One-Third Contours

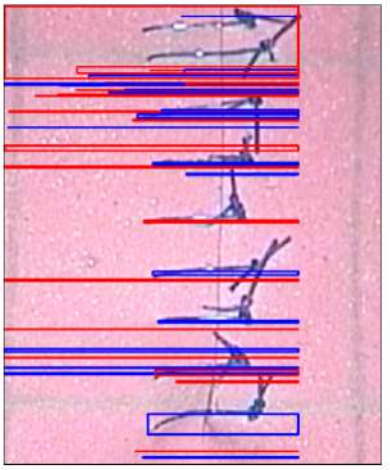
The results of One-Third Contours with respective magnification level and mAP are as follows:



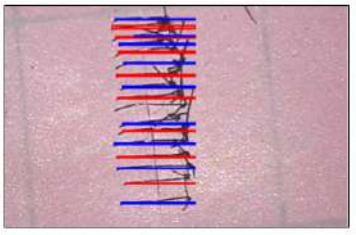
Magnification: 4



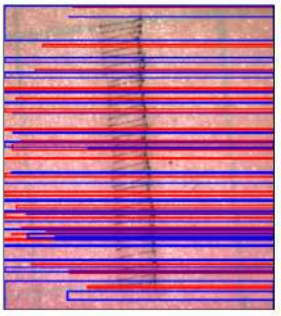
Magnification: 5



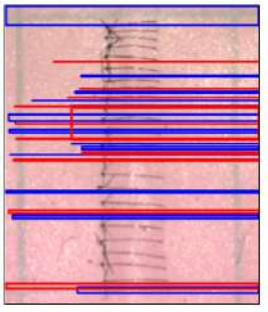
Magnification: 7



Magnification: 8



Magnification: 10



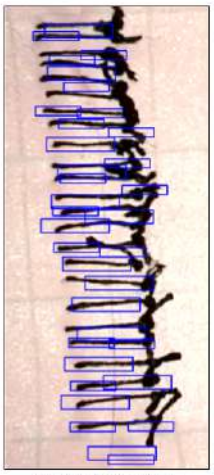
Magnification: 10

mAP for IoU of 0.3 for One-Third Contours					
Full Data-set	2.82%	For Mag of 4	2.25%	Bad Good	4.23% 0.45%
		For Mag of 5	16.83%	Bad Good	10.81% 28.33%
		For Mag of 7	0.42%	Bad Good	0.12% 1.80%
		For Mag of 8	0.00%	Bad Good	0.00% 0.00%
		For Mag of 10	1.06%	Bad Good	0.00% 4.22%

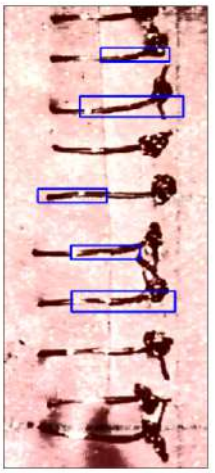
Table 4.5: Results of One-Third Contours

4.1.6 Haar Cascade

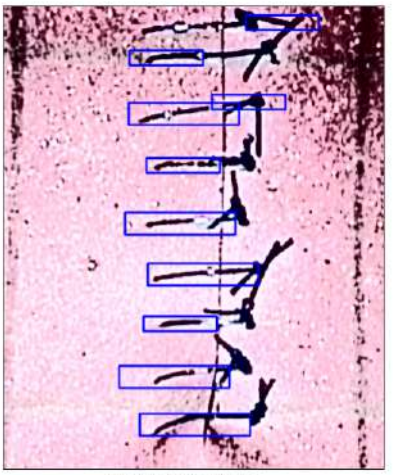
The results of Haar Cascade with respective magnification level and mAP are as follows:



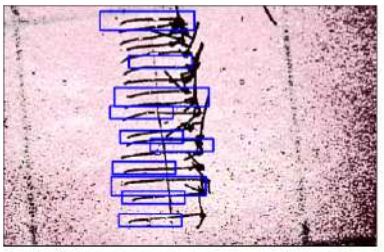
Magnification: 4



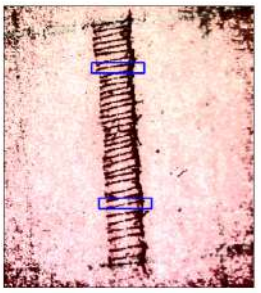
Magnification: 5



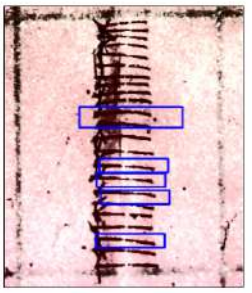
Magnification: 7



Magnification: 8



Magnification: 10



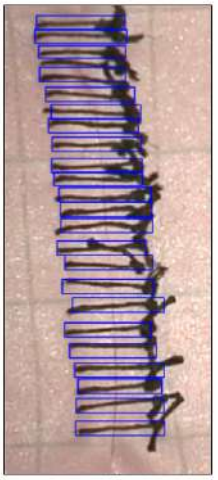
Magnification: 10

mAP for IoU of 0.3 for Haar Cascade					
Full Data-set	36.33%	For Mag of 4	51.31%	Bad Good	55.08% 45.15%
		For Mag of 5	24.53%	Bad Good	26.44% 22.82%
		For Mag of 7	44.37%	Bad Good	43.21% 51.90%
		For Mag of 8	28.10%	Bad Good	44.13% 9.22%
		For Mag of 10	29.53%	Bad Good	25.20% 32.72%

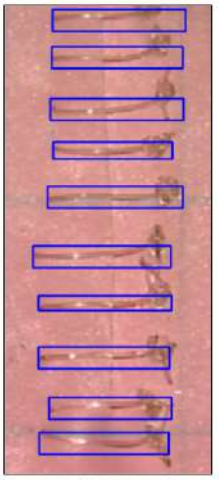
Table 4.6: Results of Haar Cascade

4.1.7 Suture Detection Using HOG-SVM

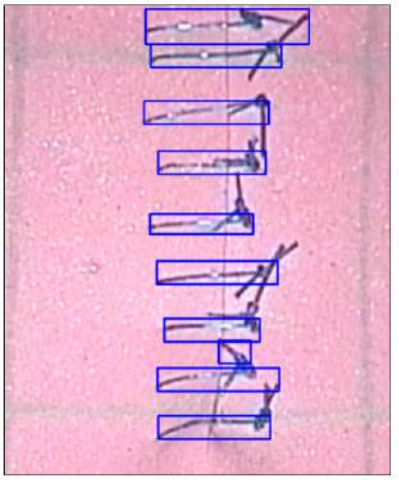
The results of HOG-SVM classifier with respective magnification level and mAP are as follows:



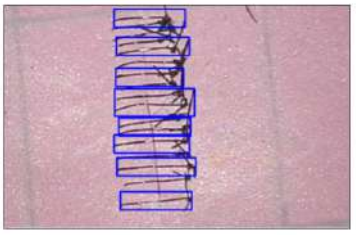
Magnification: 4



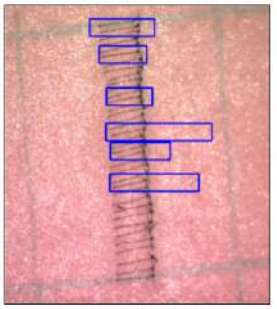
Magnification: 5



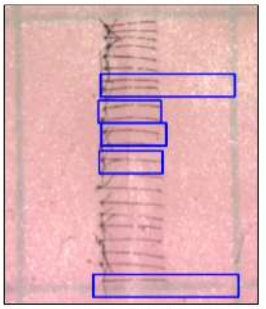
Magnification: 7



Magnification: 8



Magnification: 10



Magnification: 10

mAP for IoU of 0.3 for HOG-SVM					
Full Data-set	67.34%	For Mag of 4	89.41%	Bad Good	90.15% 94.00%
		For Mag of 5	80.66%	Bad Good	85.32% 80.54%
		For Mag of 7	77.19%	Bad Good	87.13% 80.37%
		For Mag of 8	65.46%	Bad Good	64.10% 66.04%
		For Mag of 10	29.69%	Bad Good	25.52% 32.38%

Table 4.7: Results of HOG-SVM

4.2 Parameters of Evaluation

- Uniformity in inter suture Distance - Variance in the distance between the centroids.
- Uniformity in positioning of suturing - Variance in the x value of centroids.
- Uniformity in suture size - Variance in the width of the bounding box.
- Uniformity in angulation of suture - Variance in the height of the bounding box.

Image Name	Intersuture Distance	Angulation	Size of Suture	Position of Suture
4_B_4	172.0958	65.8824	6279.5588	1801.7206
4_B_7	238.1324	40.5882	3005.8824	2295.5163
4_B_8	211.7240	41.2353	1499.9926	593.5313
4_B_10	306.1714	9.6000	386.6500	60.3500
4_G_3	144.4595	19.9958	2151.8667	650.3500
4_G_4	330.6417	50.8095	142.5714	136.6429
4_G_7	296.2518	62.7222	3907.7647	1947.1928
4_G_10	60.4437	0.8577	21.6364	815.1561
5_B_5	527.9697	279.0769	11235.1667	2650.5673
5_B_8	134.6875	13.3444	66.2667	85.2778
5_B_9	482.6292	13.1838	242.8162	116.3915
5_B_10	193.3036	56.7470	4920.9091	2284.8123
5_G_2	56.9474	23.1619	218.3905	166.6119
5_G_4	271.2026	24.8713	1513.2632	514.1199
5_G_10	582.1340	336.8421	10408.1404	3941.5965
7_B_2	111.4136	11.3333	172.6970	55.6061
7_B_4	90.6227	38.2045	1438.0227	287.0966
7_B_5	381.0278	28.6222	2009.8778	164.1139
7_B_8	551.1625	112.9412	6915.4412	4640.1581
7_G_2	206.4967	25.4503	989.9181	8501.3304
7_G_4	49.4242	0.0000	400.8333	422.5353
7_G_10	143.6319	103.6556	2256.9889	475.4472
8_B_1	90.9524	32.0000	50.4107	20.8884
8_B_6	167.0074	0.0000	504.5000	5803.0074
8_G_2	224.6838	64.4183	5308.0294	1767.9943
8_G_5	172.8529	25.4503	663.8772	774.5833
10_B_5	238.7000	0.0000	1816.4000	801.4667
10_B_6	87.9861	14.2333	347.7333	37.2778
10_G_1	4497.6667	0.0000	3284.5000	708.0750
10_G_3	211.9459	93.2921	809.9237	3340.7178

Figure 4.1: Parameters of Evaluation

The above table shows all the values of evaluation parameters for all the images in test dataset.

Chapter 5

Conclusions and future scope

5.1 Conclusion

In this project, from the images of suturing performed by trainee neurosurgeons, we have explored different methods for detection and localization of the sutures. From the above results we can conclude that, the techniques like PHough, adaptive thresholding and contours worked fine only for few cases. But these techniques can't be generalised as there are many hyper parameters to be tuned for individual images.

Hence, to generalize, we implemented the classical object detection techniques to detect the sutures. From the classical object detection technique, Viola-Jones did not perform well in detecting the sutures. The Opencv Haar Cascade classifier technique, gave us better results as compared to Viola-Jones, to improve the performance of the classifier, we introduced the LBP (Local Binary Pattern) feature. The LBP feature classifier gave us better results. However, HOG-SVM approach gave us good results compared to all the methods we tried. So, we opted this approach for suture detection.

Based on these detections, we were able to provide scores for various parameters like inter-suture distance, angulation in the suture,size of the suture and position of the suture. These scores are further combined to provide effectualness.

Hence, we conclude that the normal image processing and vision features alone did not fetch us good results due to cumbersome tuning of parameters for every individual image and needed generalization. Whereas the generalized basic vision-based learning techniques:

- Fetched us better results
- Detected most of the sutures with appropriate aspect ratio with the suture

5.2 Future Scope

System efficiency analysis i.e, to keep the system time complexity as low as possible by maintaining conditional independence between the parameters.

Bibliography

[1] J. Cruz, E. Shiguemori, and Lamartine Guimarães. A comparison of haar-like, lbp and hog approaches to concrete and asphalt runway detection in high resolution imagery. *Journal of Computational Interdisciplinary Sciences*, 6, 01 2016.

[2] N. Dalal and B. Triggs. Histograms of oriented gradients for human detection. In *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05)*, volume 1, pages 886–893 vol. 1, 2005.

[3] P. Viola and M. Jones. Rapid object detection using a boosted cascade of simple features. In *Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition. CVPR 2001*, volume 1, pages I–I, 2001.